

Self-Sensing GFRP / CNT Composites under Flexural Loading

Failure Forecasting from Electrical Resistance

Priyanshu Kumar (AE22B048)

Guide: Prof. David Kumar

Department of Aerospace Engineering, IIT Madras

Code and report: github.com/PriyanshuKumar9798/self-sensing-gfrp-cnt

AS4545 | Undergraduate Research Project



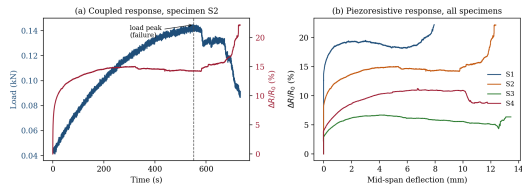
Motivation

- Composites carry load in critical structures such as aircraft panels and wind-turbine blades.
- Failure usually begins *inside* the material, as micro-cracks, where it is invisible.
- Detecting it means taking the structure out of service to inspect: slow and costly.
- A self-sensing material would report its own condition electrically, while still in use.

“Damage starts where you cannot see it. We want the material to tell us before it breaks.”

Sensing mechanism

- Glass fibres do not conduct; the carbon nanotubes (CNTs) form a conducting network in the matrix.
- Strain widens tunnelling gaps and micro-cracks sever paths, so resistance rises.
- The rise is *exponentially* sensitive, so even tiny early damage can show up.
- The signal reports on the matrix, exactly where damage starts.



Objective

Predict, from the resistance signal alone, how close a bar is to breaking, and raise an early-warning alarm, on bars the model has never seen.

Forecast

Estimate remaining
deflection to failure

Warn

Trigger a calibrated alarm
before failure

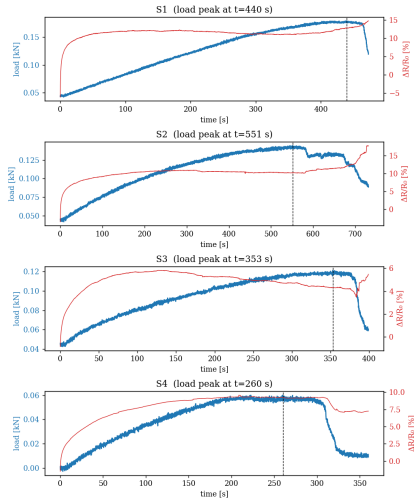
Generalise

to a new, unseen bar

Experimental setup and data

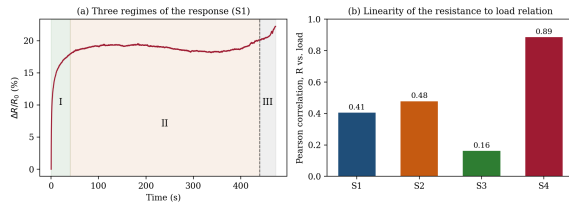
- Four bars (S1–S4) bent to failure in three-point bending.
- A universal testing machine recorded load and deflection; a Keithley source-meter recorded resistance.
- Both at ~ 12 Hz on a common time base; resistance used as the fractional change $\Delta R/R_0$.

Bar	$R_0(\text{M}\Omega)$	Peak(kN)	$D_{\text{peak}}(\text{mm})$	Rate(mm/s)
S1	52.4	0.180	7.36	0.017
S2	61.8	0.144	9.26	0.017
S3	11.1	0.121	11.55	0.033
S4	6.8	0.060	8.44	0.033



Resistance response: three regimes

- **I** Steep early rise: most of the change in the first 10–20% of the test.
- **II** Saturation: a plateau through the middle while load keeps climbing.
- **III** Features near failure: clear steps and jumps around the load peak.
- Resistance–load correlation varies 0.16 to 0.89: no fixed mapping holds across bars.



Challenges and mitigations

Only four bars, so models overfit → Leave-one-bar-out testing, simple models first, and heavy data augmentation

Loading rate and baseline resistance differ between bars → Rate-invariant target (deflection-to-failure); work in $\Delta R/R_0$

Intervals over-confident (90% band covered only 61%) → Cross-conformal correction to restore valid coverage

Methodology



Deflection to failure (the target): how much further the bar can bend before it breaks. It does not depend on the loading speed, so it is comparable across bars.

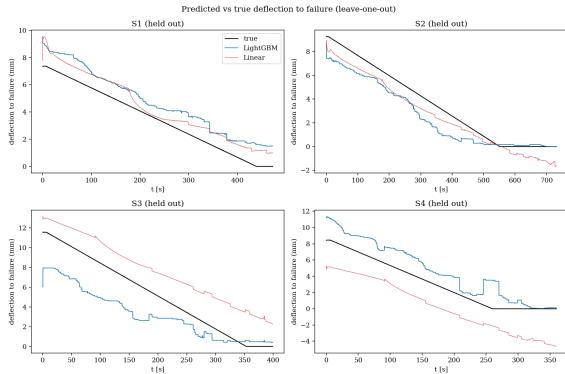
Leave-one-specimen-out (the test): train on three bars, test on the fourth the model has never seen, then rotate. This is the honest measure of accuracy on a new bar.

Results: classical baselines

- Gradient-boosted trees (LightGBM) with a linear-regression floor.
- Tuned and tested entirely under leave-one-specimen-out.
- Predictions track the falling line to failure on every held-out bar.

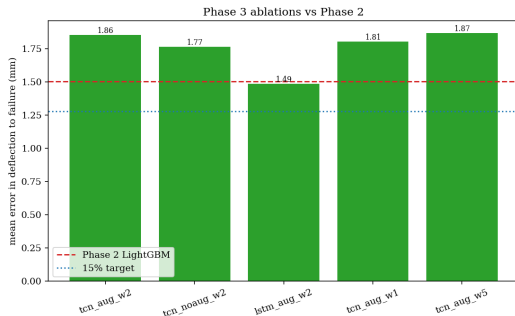
1.51 mm

mean deflection-to-failure error
(LightGBM)



Results: sequence models

- Small TCN and LSTM networks with five physics-aware augmentations.
- Best sequence model: LSTM at 1.49 mm, essentially level with the baseline.
- No model reached the 15% improvement target.
- With four bars, gradient-boosted trees remain competitive, as expected.

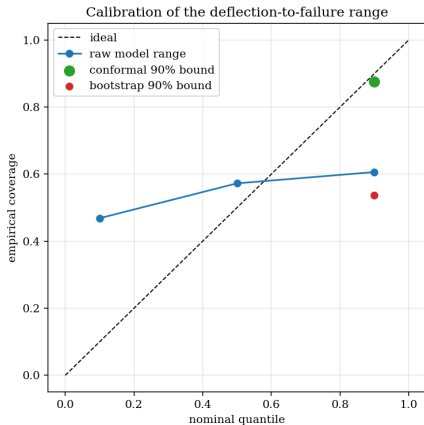


Results: uncertainty calibration

- The model predicts a range (10th to 90th percentile), not a single number.
- Within one bar the deflection to failure is almost fixed, so the raw range came out too narrow and over-confident.
- A conformal correction widens the bound until its stated 90% coverage actually holds.

0.61 → 0.88

coverage of the 90% bound, before and after
calibration

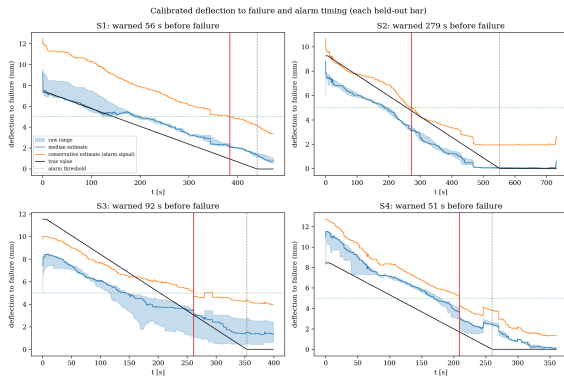


Results: early-warning alarm

On every held-out bar, the alarm fires *before* failure, with no false alarms.

1.00 **1.00** **74 s**
precision recall median warning

Operating point: raise the alarm when the conservative deflection-to-failure bound stays below 5 mm for 3 samples in a row.



Deployment and validation

Predicting on a new bar, in real time

Live signal: resistance + deflection → causal features (past only) → trained model → deflection to failure, then warning in seconds

It uses only the past, so it can run while the bar is still being loaded.

Checking the accuracy

- Load a new bar to failure: its load peak fixes the true remaining deflection at every instant.
- Compare the prediction against that truth, giving the average error in millimetres, exactly as measured on the four bars.
- Each additional bar is one more independent test; the current honest estimate is the leave-one-out error of about 1.5 mm.

Run it on a new bar: `make predict UTM=newbar.xls SM=newbar.csv`

Limitations

- Only four specimens, so every reported number has a wide confidence band.
- Two loading rates and an eightfold baseline-resistance spread remain real confounds.
- A two-wire measurement includes contact resistance.
- One material and one geometry: the results do not yet generalise further.

Conclusions and future work

- A model that reads the resistance signal predicts deflection-to-failure of an unseen bar to about 1.5 mm.
- A conformally-calibrated alarm warns before failure on every held-out bar, about 74 seconds ahead, with no false alarms.
- Sequence models did not beat the simpler trees, an honest reflection of having only four bars.

Next step. Test more specimens, ideally with a four-wire measurement, to tighten every estimate and give the sequence models a fair chance.

Thank you

Questions and discussion

Priyanshu Kumar | AE22B048 | Department of Aerospace Engineering, IIT Madras

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